My white board writing on 1st August

① I illustrated the spectral decomposition of a real symmetric matrix:

Let $X = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$. Find the spectral decomposition.

First we find the eigenvalues: $|X-\lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 2 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda)-4 =$
$= \lambda^2-4\lambda-1 = 0$

Get:
$\lambda_1 = 2+\sqrt{5}$, $\lambda_2 = 2-\sqrt{5}$.

Next, we find the normed eigenvector that corresponds to λ_1 by solving first:

$$X\begin{pmatrix} a \\ b \end{pmatrix} = (2+\sqrt{5})\begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow$$ This equation system has many solutions (we know that the eigenvector is not uniquely defined. The first equation reads: $a+2b = (2+\sqrt{5})a$, hence $2b-(1+\sqrt{5})a = 0$. If we set $b=1$ we get $a = \frac{2}{1+\sqrt{5}} \approx .618$. This eigenvector is not normed. We norm it to get

$$e_1 = \begin{pmatrix} \frac{.618}{\sqrt{1+.618^2}} \\ \frac{1}{\sqrt{1+.618^2}} \end{pmatrix} = \begin{pmatrix} .5257 \\ .8506 \end{pmatrix}$$ (up to round-off)

In a similar way, when solving $X\begin{pmatrix} c \\ d \end{pmatrix} = (2-\sqrt{5})\begin{pmatrix} c \\ d \end{pmatrix}$,

the first equation gives $2d-(1-\sqrt{5})c = 0$. If we set $c=1$, we get $d = \frac{1-\sqrt{5}}{2} = -0.618$. By norming it we get

$$e_2 = \begin{pmatrix} .8506 \\ -.5257 \end{pmatrix}$$

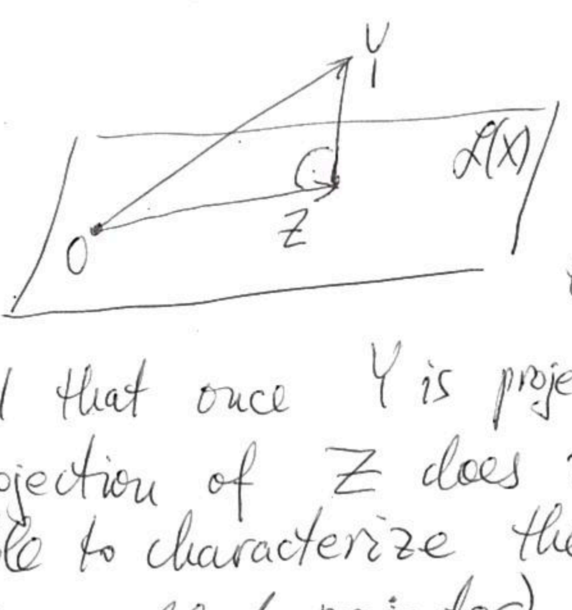
[Note that we could have got e_2 also directly after finding e_1) by using the fact that $(e_1, e_2) = 0$ must hold!)

Hence $$X = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = (2+\sqrt{5})\begin{pmatrix} .5257 \\ .8506 \end{pmatrix}(.5257 \quad .8506) + (2-\sqrt{5})\begin{pmatrix} .8506 \\ -.5257 \end{pmatrix}(.8506 \quad -.5257)$$

Next, I noticed explicitly that the above example covers the case where the matrix X was not positive definite (yet the spectral decomposition above is valid!) For positive definite matrix, all $\lambda_i > 0$, $i=1,2,\dots,p$ holds. For non-negative definite matrix, all $\lambda_i \geq 0$, $i=1,2,\dots,p$ holds. If the matrix is non-negative definite then we can define its square-root in the following way: if $X = \sum_{i=1}^{p} \lambda_i e_i e_i'$ then we define

$X^{\frac{1}{2}} = \sum_{i=1}^{p} \sqrt{\lambda_i}\, e_i e_i'$ and we check directly that, due to the orthonormality of the vectors e_i, $i=1,2,\dots,p$, we have $X^{\frac{1}{2}} X^{\frac{1}{2}} = X$.

2) I then discussed orthogonal projection of arbitrary vector $Y \in R^n$ on the p-dim linear subspace spanned, say, by the columns of a matrix $X \in \mathcal{M}_{n,p}$.



$Z = PY$. Using the geometric facts that $Y-Z$ is perpendicular to any vector in $\mathcal{L}(X)$

and that once Y is projected once, giving Z, the projection of Z does not change Z, we were able to characterize the projection matrix P (the so-called projector) as symmetric and idempotent. All the detailed discussion are in the notes and I abstain from reproducing them again here

I gave you an exercise to show that the rank of an orthogonal projector is equal to the sum of its diagonal elements. I gave you a hint:

(i) Show that an eigenvalue of a projector can only be zero or one.

(ii) Use the spectral decomposition and apply the trace operation, using properties of traces.

I then started the second lecture. I did not have to write much on the white board since the discussion in the notes is thorough enough. Perhaps I could elaborate on the proof of Property 3 on MVN which I mentioned without writing down on the white board:

Property 3: $X \sim N_p(\mu, \Sigma)$ is subdivided into subvectors $X = \begin{pmatrix} X_{(1)} \\ X_{(2)} \end{pmatrix}$ and according to this subdivision we subdivide $\mu = \begin{pmatrix} \mu_{(1)} \\ \mu_{(2)} \end{pmatrix}$ and $\Sigma = \left(\begin{array}{c|c} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{array}\right)$. Then the vectors $X_{(1)}$ and $X_{(2)}$ are independent if and only if (iff) $\Sigma_{12} = 0$.

Proof: ($\Longrightarrow$) If the vectors are independent they are uncorrelated anyway since $\mathrm{cov}(\tilde{X}, \tilde{Y}) = E(\tilde{X}\tilde{Y}) - E\tilde{X}.E\tilde{Y} = E\tilde{X}.E\tilde{Y} - E\tilde{X}.E\tilde{Y} = 0$ holds for any two random variables $\tilde{X}, \tilde{Y}$. Because of the independence. Hence $\Sigma_{12} = 0$.

($\Longleftarrow$) The more interesting part. Let us investigate the joint characteristic function:

$$\varphi_X(t) = \varphi_{\binom{X_{(1)}}{X_{(2)}}}(t_{(1)}, t_{(2)}) = e^{it'\mu - \frac{1}{2}t'\Sigma t} =$$

$$\overset{\uparrow}{=} e^{it'_{(1)}\mu_{(1)} + it'_{(2)}\mu_{(2)} - \frac{1}{2}t'_{(1)}\Sigma_{11}t_{(1)} - \frac{1}{2}t'_{(2)}\Sigma_{22}t_{(2)}}$$

Because of $\Sigma_{12} = 0$

$$= e^{it'_{(1)}\mu_{(1)} - \frac{1}{2}t'_{(1)}\Sigma_{11}t_{(1)}} * e^{it'_{(2)}\mu_{(2)} - \frac{1}{2}t'_{(2)}\Sigma_{22}t_{(2)}}$$

But $\uparrow$ this is just the product of the two characteristic functions of $N(\mu_{(1)}, \Sigma_{11})$ and $N(\mu_{(2)}, \Sigma_{22})$. That is, the joint characteristic function is decomposed into a product form. Hence $X_{(1)}$ and $X_{(2)}$ are <u>independent vectors.</u>

I finished the lecture by formulating <u>property 4</u>. Since it is very important for further discussions, I left its proof for the beginning of next week's lecture.